SEMINAR SUMMARY:
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NUMERICAL SOLUTIONS OF BIOLOGICALLY BASED PROBLEMS USING DIFFERENTIAL EQUATIONS

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In general across all the projects we explore below in this summary we have a general framework around how one of these projects is structured. Specifically the goal is to collaborate with biologist to understand research questions and the current state of the field. From this collaboration, our goal is to develop mathematical models that accommodate using numerical techniques for solving them. We use the biological data to paramaterize the model and/or run a sweep of the parameter space. Though it is worth noting that there is a natural trade-off between mathematical complexity of the model and the solvability of the model.

The projects that we examined specifically were:

- Wound healing and inflamation
- Wound healing in coral
- Gut health and inflammation
- Nematocyst firing
- Phage therapy
- Sickle Cell Pain Episode Prediction

Our summaries on projects 5. and 6. were generally short because those both were described i n the last five-minutes of class.

1. Wound healing and inflammation.

For this project the original data came from an experiment that the original biologist did on himself (with medical assistance). Spefically he had a "PTFE" (or vortex) tube inserted under his skin for a period of time to create the wound in question as a mechanism for creating a standardized wound. Not surprisingly, there were few other participants in the clinical trial. That said, because it was biologist himself doing the experiment the richness of the data is interesting. For example, that there were even mounted slides from the experiment that showed the PTFE tube matrix along with fibroblasts laying down collagen.

The model in question was a partial differential equation model based on the variables m for the density of damaged the tissue, n for the activated inflammatory cells, f for the

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fibroblasts, and c for new collagen deposition. We have four model equations, starting with the first for phagocytosis:

$$\frac{\partial m}{\partial t} = -\mu_{mn} mn$$

The second equation represented diffusion and chemotaxis and deactivation, given by:

$$\frac{\partial n}{\partial t} = \nabla \cdot [D_n \nabla n - n\chi_n \nabla m] - \mu_{mn} mn$$

The third equation represented diffustion/motility and chemotaxis and cell division and cell death

$$\frac{\partial f}{\partial t} = \nabla \cdot [D_f \nabla f - f\chi_f \nabla n] + s_f(n) f \left(2 - \frac{f}{F_0}\right) - \mu_f f$$

And the fourth model equation represents deposition by f and collateral damage

$$\frac{\partial c}{\partial t} = k_c f (C_0 - c) - \mu_{cn} cn.$$

We were then presented some plots that represented collagen and fibroblast levels versus days with modeled data and real data ploted on the graphs.

2. Wound healing in coral.

The goal in this project is to consider coral in place of human tissue. Biologically we know that the damage processes (laceration, avulsion, fracture), inflammation processes, and physiological process for tissue regrowth behave in a similar fashion to humans. We noted that it was a convenient wound healing mechanism to study because the results could indeed be produced with coral in the lab as well as naturally occurring coral. We ask questions like whether the wound is regrowing or propagating. There were two specefied aims of the project, namely cellular level-wound healing versus tissue level-enegery focused

In both of the aims of the project we began examining whether coral bleaching had any effect on the speed and mechanics of wound healing. Though it is know that glycerol production as a photosyntehtic byproduct transferred to mucus from the symbiotic algae yielded energy transfer to the calioblastic cells for skeletal growth.

Potential future research projects out of this body of work could be examining multiple stressors, other coral species and linked cellular and tissue level models.

3. Gut health and inflammation.

So it's known that gut health has a considerable amount to do with body health from general inflammation, to mental health, to even levels of atherosclerosis. Now gut inflammation is regulated by an enzyme called alkaline phosphate (specifically intestinal alkaline phosphate or IAP). And, the gram-negative bacteria in the colon produce lipopolysaccharides (LPS) which are inflammatory agents. So the interplay between IAP and LPS regulates how the gut performs and even the structural integrity of wall of the bowel.

If I understand the slides correctly our goal here is to write a mathematical model that effectively encompases the metabolic pathway for the breakdown of LPS by IAP, which was given in a diagram in one of the slides. The equations that arise from this metabolic pathway follow as:

$$\begin{aligned}
\frac{dIAP}{dt} &= \alpha S_I - \rho_I IAP * LPS_G - \mu_I I \\
\frac{dP}{dt} &= \theta_I \left(\frac{LPS_G}{X_{ad} + LPS_G} \right) - \theta_R \frac{P}{P + K_R} \\
\frac{dLPS_G}{dt} &= \beta S_l - \rho_L IAP * LPS_G - P(LPS_G - LPS_B) \\
\frac{dLPS_B}{dt} &= DP(LPS_G - LPS_B) - \mu_L LPS_B
\end{aligned}$$

After getting these equations we saw a variety of plots. The first plots were actual animal data taken from lab mice. We looked at the differences between a “chow” diet and a “western” diet over a multiple week period and how inflammation occurred in the animal. Further we saw how changes in diet could affect inflammation. Lastly, we were presented with a variety of numerical plots of data generated from the model comparing a healthy diet, a continued “western” diet a partially healthy diet and then those with stimulus where there was a stimulus to affect the production of IAP in a positive manner.

4. Nematocyst firing.

5. Phage therapy.

6. Sickle cell pain episode prediction.